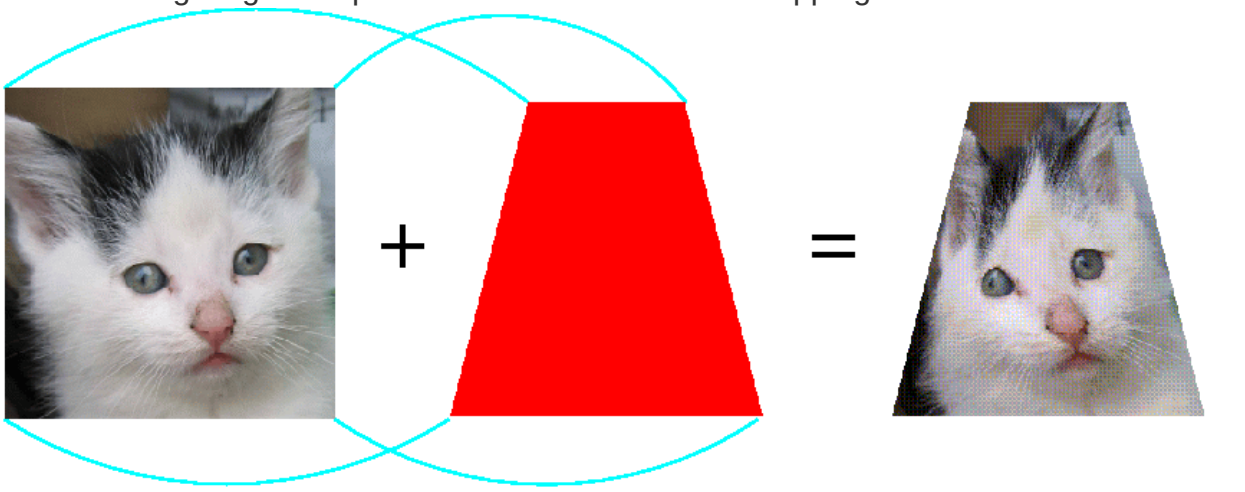
CS3110- Introduction to Computer Graphics
Lab Assignment 3
Texture Mapping and Multi-Texturing
Fall 2026

Lab Notes

Overview and Definitions

The following diagram represents the idea of texture mapping:



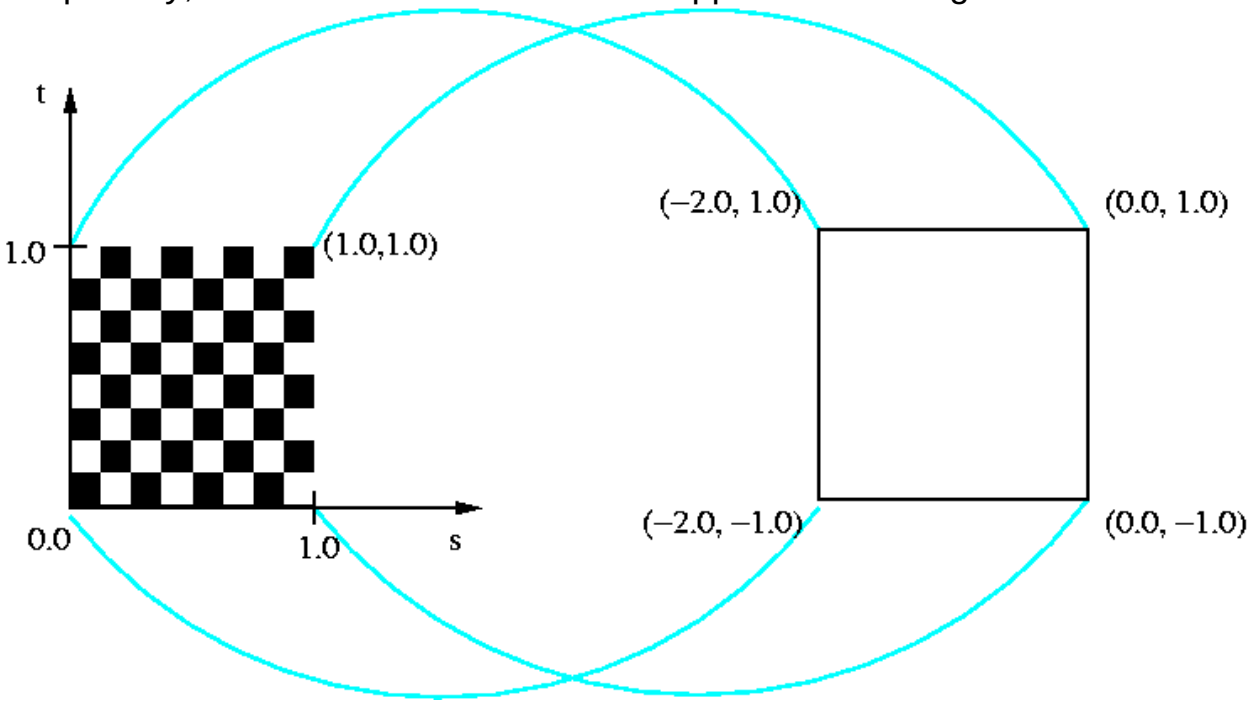
Some definitions:

- **Texture Mapping**--"a method of adding realism to a computer-generated graphic. An image (the texture) is added (mapped) to a simpler shape that is generated in the scene, like a decal pasted to a flat surface. This reduces the amount of computing needed to create the shapes and textures in the scene. For instance, a sphere may be

generated and a face texture mapped, to remove the need for processing the shape of the nose and eyes."

- **Textures**--rectangular arrays of data. The data can be color data, luminance data, or color and alpha data
- **Texels**--the individual values in a texture array
- **Mipmaps**--Although we will not be covering this in detail in this lab. Mipmaps are often discussed with texture mapping. It's something good to know. The idea is that you have multiple images to cover multiple levels of detail. For instance, for an object far away, you have a texture or a mipmap that is small and with little detail, and for closer objects, you have a texture that is large and detailed.

Graphically, the texture coordinates are mapped as following

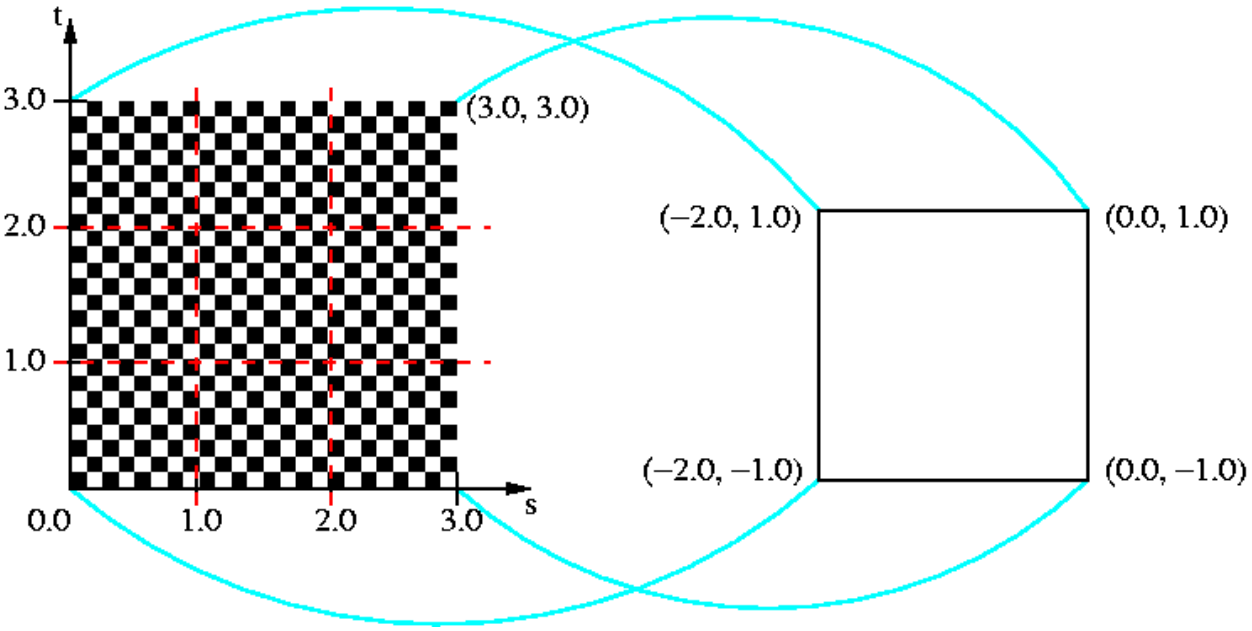


Repeating a Texture Pattern or Clamping it

In the example that we are using, we have specified that we would like to repeat the pattern, but we are currently not making use of this repeating feature. When we repeat a texture, it provides a "tile" effect. The following code specifies that we will repeat the texture map.

```
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_WRAP_S, GL_REPEAT);
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_WRAP_T, GL_REPEAT);
```

Graphically, we could see the mapping as the following (the red lines are placed in the texture map to help show that the texture is repeated three times)



Goal of Assignment

There are two main objectives of this second assignment in CS3110. The first is to learn how basic transformations work by applying them to a simple two-dimensional shape. The second is to explore a variety manipulation that can be applied to the samples (pixels) of an image to achieve different effects.

Task 1- Extend the Sperinski Triangles from last assignment and add a texture of **your own picture** in every triangle. Also use different filtering options on each triangle (such as clamping, mirroring and repeat).

Task 2- Extends the Sperinski Triangles and **add a different texture (Multi Texturing)** to each triangle. The images file will be provided with this assignment

Deliverables

1. Your code + any supporting files (for instance, if you put custom functions or points arrays in separate .js files)
2. An image of your final product